

Integration over a Vanishing Cycle

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Introduction

Let k be a commutative field and let f be a formal power series in two indeterminates x and y over k :

$$f = \sum a_{n,m} x^n y^m.$$

Let $I(f)$ be the formal power series in one indeterminate t

$$I(f) = \sum a_{n,n} t^n.$$

We propose to prove the following result:

Theorem. *If f represents an algebraic function of x and y , and if k has characteristic $p \neq 0$, then $I(f)$ represents an algebraic function of t .*

For g a formal power series in N indeterminates

$$g = \sum a_\nu x^\nu,$$

one defines similarly

$$I_N(g) = \sum_n a_{n,1} t^n,$$

and one again has:

Corollary. *If g is algebraic over the field $k(x_1, \dots, x_N)$, and if k has characteristic $p > 0$, then $I_N(g)$ is algebraic over $k(t)$.*

Under the stronger hypothesis $g \in k(x_1, \dots, x_N)$, this result is due to H. Furstenberg [1]. See also 3.10. Our first aim was to give a geometric explanation of Furstenberg's result.

Suppose first that $k = \mathbb{C}$ and that the formal series f is convergent. Then $I(f)$ is convergent, and admits the following integral representation

$$I(f)(t) = \frac{1}{2\pi i} \oint_{xy=t} f(x, y) \frac{dx \wedge dy}{dt}. \quad (1)$$

Let us explain the meaning of the right-hand side:

a) Let $\pi : \mathbb{C}^2 \rightarrow \mathbb{C}$ be the map $(x, y) \mapsto xy$. The expression $(dx \wedge dy)/dt$ is a relative holomorphic differential form on $\mathbb{C}^2 - \{(0, 0)\}$, relative to the projection π , i.e. a family of holomorphic differential forms on the fibres, depending holomorphically on t . At $\{(0, 0)\}$, it is still a section of the relative dualizing

sheaf ω , defined in the duality formalism of A. Grothendieck by $R\pi^!\mathcal{O} = \omega[1]$. In terms of the coordinate y on $\pi^{-1}(t)$, one has

$$\frac{dx \wedge dy}{dt} = \frac{d(t/y) \wedge dy}{dt} = \left(\frac{dt}{y} \wedge dy \right) / dt = \frac{dy}{y}.$$

b) Suppose f converges in the polydisc B of radii R . For $|t|$ sufficiently small ($|t| < R^2$), the intersection of B with $\pi^{-1}(t)$ projects isomorphically in the y -plane onto the annulus $|t|/R < |y| < R$. Let r be such that $|t|/R < r < R$. The integral $\oint_{xy=t}$ is taken over the cycle $|y| = r$, $|x| = |t|/r$, $xy = t$, which projects isomorphically onto the positively oriented circumference $|y| = r$; it is independent of the choice of r .

These explanations show that

$$\frac{1}{2\pi i} \oint_{xy=t} f(x, y) \frac{dx \wedge dy}{dt} = \frac{1}{2\pi i} \oint f(t/y, y) \frac{dy}{y} = \frac{1}{2\pi i} \oint \sum a_{n,m} t^n y^{m-n} \frac{dy}{y}.$$

To deduce (1), one exchanges summation and integration, and applies the residue formula: only the terms for which $m = n$ remain.

The integration cycle is the vanishing cycle, in the sense of Lefschetz, for π , near $(0, 0)$. If one takes $r = |t|^{1/2}$, one indeed sees it disappear as $|t| \rightarrow 0$. This explains the title of the article.

In the definition of $I(f)$, the indeterminates x and y play symmetric roles. In (1), an order between x and y determines two choices of signs, which cancel one another: in the relative differential form, and in the orientation of the vanishing cycle.

Formula (1) suggests that the operation I has an intrinsic meaning in the following setting: start from a morphism of formal schemes $\pi : X \rightarrow S$ over k , with X isomorphic to $\mathrm{Spf} k[[x, y]]$, S isomorphic to $\mathrm{Spf} k[[t]]$, and π having a non-degenerate quadratic singularity. Assume that the two branches of $\pi^{-1}(0)$ are defined over k , and choose an order between them. In other words, assume that there are formal coordinate systems x, y on X and t on S in which π is $(x, y) \mapsto xy$; these coordinate systems will always be taken so that the first branch of $\pi^{-1}(0) : xy = 0$ is the branch $x = 0$.

Let $\Omega_{X/k}^1$ be the Kähler differentials, defined while taking the topology into account: if $X = \mathrm{Spf}(A)$, with A isomorphic to $k[[x, y]]$ and endowed with the (x, y) -adic topology, then $\Omega_{X/k}^1$ corresponds to the A -module $\varprojlim \Omega_{(A/I)/k}^1$, the projective limit over the open ideals. Similarly for S . In coordinates, $\Omega_{X/k}^1$ is the free $\mathcal{O}_X$ -module with basis dx, dy , and $\Omega_{S/k}^1$ is the free $\mathcal{O}_S$ -module with basis dt . For a free module M of rank n , put $\det M = \bigwedge^n M$. For L free of rank 1, denote by L^{-1} its dual. Put $\omega_{X/k} = \det \Omega_{X/k}^1$, $\omega_{S/k} = \Omega_{S/k}^1$, and let $\omega_{X/S}$ be the relative dualizing sheaf

$$\omega_{X/S} = \pi^! \mathcal{O}_S[1] = \omega_{X/k} \otimes \pi^* \omega_{S/k}^{-1}.$$

In coordinates, it is the free $\mathcal{O}_X$ -module with basis $dx \wedge dy/dt$. We will interpret I as a morphism

$$\pi_* \omega_{X/S} \longrightarrow \mathcal{O}_S,$$

i.e., for $S = \mathrm{Spf}(V)$, as a V -linear morphism

$$H^0(X, \omega_{X/S}) \longrightarrow V.$$

Section 1 gives this interpretation: there we prove, for arbitrary k , a formula which is an algebraic analogue of (1). The integration sign which appears in (1) is replaced by Grothendieck's local trace morphism. More precisely, let t be a uniformizer of S (for instance t for $S = \mathrm{Spf}(k[[t]])$). We will write a subscript n for reduction modulo t^n . We omit the subscript when there is no risk of confusion. For example, we will still write $\pi : X_n \rightarrow S_n$ for the morphism π_n deduced from π by reduction modulo t^n . We write ω for the inverse image $(\omega_{X/S})_n$ of $\omega_{X/S}$ on X_n . This is Grothendieck's relative dualizing sheaf ω_{X_n/S_n} . Let A_n be the complete local ring whose formal spectrum is X_n ; in coordinates, $A_n = k[[x, y]]/((xy)^n)$. It will be convenient to work not with the formal scheme X_n , but with the ordinary scheme $X'_n := \mathrm{Spec} A_n$. The difference is Byzantine: both X_n and X'_n are merely geometric ways of speaking about A_n . Let O be the closed point of X'_n ; in coordinates, $x = y = 0$. Grothendieck's local trace morphism is a morphism

$$\mathrm{Tr} : H^1_{\{O\}}(X'_n, \omega) \longrightarrow H^0(S_n, \mathcal{O}).$$

One has

$$H^1_{\{O\}}(X'_n, \omega) = \varinjlim \mathrm{Ext}^1_{A_n}(A_n/I, \omega),$$

the inductive limit being over the open ideals I of A_n . It is to write this inductive limit as a cohomology group with support that we had to introduce X'_n .

The role of the integration cycle, the vanishing cycle, will be played by an element $v_n \in H^1_{\{O\}}(X'_n, \mathcal{O})$. In coordinates, this element lies in the image of $\mathrm{Ext}^1_{A_n}(A_n/(x+y)^{2n-1}, A_n)$. Note that the required exponent of $(x+y)$ grows with n .

In Section 2 we consider a scheme in curves Y which is proper and flat over S , from which X is obtained by completion at a point, and we relate the v_n to an element of $\mathrm{Lie} \mathrm{Pic}_S Y$. In Section 3 we globalize X and S , and prove the theorem and the corollary.

Notation. We will write base changes as subscripts. For example, if X and T are S -schemes, $X_T = X \times_S T$. Reduction modulo t^n is denoted by a subscript n .

For our purposes, the sign conventions used are immaterial. For the intermediate results stated below to be true as written – and not only up to sign – one must nevertheless choose them. Our choice is made explicit in the appendix.

1 Algebraic analogue of (1)

1.1.

Keep the notation of the introduction, and let x, y and t be coordinates on X and S of the type already considered.

The inclusion of X'_1 into X'_n is a homeomorphism, since X'_1 is defined in X'_n by a nilpotent ideal. Thus the complement of the origin $\{O\}$ in X'_n has two connected components, corresponding to the two branches $x = 0$ and $y = 0$ of X'_1 . Let u_n be the section of the structure sheaf of $X_n - \{O\}$ which has

value 1 on the first branch and 0 on the second. In the language of rings, the homomorphism

$$A_n = k[[x, y]]/((xy)^n) \longrightarrow k[[x, y]]/(x^n) \times k[[x, y]]/(y^n)$$

induces an isomorphism

$$A_n((x + y)^{-1}) \xrightarrow{\sim} (k[[x]]/(x^n))((y)) \times (k[[y]]/(y^n))((x)),$$

and $u_n \in A_n((x + y)^{-1})$ has image $(1, 0)$. Explicitly,

$$u_n = \left[\sum_{i=0}^{n-1} \binom{2n-i}{i} y^{2n-1-i} x^i \right] / (x + y)^{2n-1}. \quad (1.1.1)$$

Define v_n to be the image of u_n by the natural boundary morphism

$$H^0(X'_n - \{O\}, \mathcal{O}) \longrightarrow H^1_{\{O\}}(X'_n, \mathcal{O}).$$

The long exact sequence of the $\text{Ext}^i(*, A_n)$, applied to the short exact sequence

$$0 \longrightarrow ((x + y)^N) \longrightarrow A_n \longrightarrow A_n/((x + y)^N) \longrightarrow 0$$

gives a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_n & \longrightarrow & (x + y)^{-N} A_n & \longrightarrow & \text{Ext}_{A_n}^1(A_n/((x + y)^N), A_n) \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H^0(X'_n, \mathcal{O}) & \longrightarrow & H^0(X'_n - \{O\}, \mathcal{O}) & \longrightarrow & H^1_{\{O\}}(X'_n, \mathcal{O}). \end{array}$$

For $N = 2n - 1$, this diagram and (1.1.1) show that v_n lies in the image of $\text{Ext}_{A_n}^1(A_n/((x + y)^{2n-1}), A_n)$.

1.2. Proposition.

Let f belong to $k[[x, y]]$. For every n , one has

$$\text{Tr}(f dx \wedge dy/dt \cdot v_n) = -I(f) \pmod{t^n}. \quad (1.2.1)$$

In (1.2.1), Tr is Grothendieck's local trace morphism. It is applied to the element of $H^1_{\{O\}}(X'_n, \omega)$ obtained as the product of $f dx \wedge dy/dt$ in $H^0(X'_n, \omega)$ and v_n in $H^1_{\{O\}}(X'_n, \mathcal{O})$. We prove (1.2.1) under the hypothesis that k is infinite. The general case is reduced to this one by extension of scalars.

Let $I'(f)$ denote the left-hand side of (1.2.1). We have $I'(f) \in k[[t]]/(t^n)$. If J is an open ideal of A_n , small enough that v_n lies in the image of $\text{Ext}^1(A_n/J, A_n)$ (for example $J = ((x + y)^{2n-1})$), and if $f \in J$, then $f v_n = 0$, and *a fortiori* $I'(f) = 0$: the $k[[t]]/(t^n)$ -linear map

$$I' : A_n \longrightarrow k[[t]]/(t^n), \quad f \longmapsto I'(f)$$

is continuous.

For $\lambda, \mu \in k$, let $\sigma(\lambda, \mu)$ be the automorphism $(x, y) \mapsto (\lambda x, \mu y)$, $t \mapsto \lambda \mu t$ of (X, S, π) . It preserves $dx \wedge dy/dt$ and v_n . By transport of structure, one therefore has

$$I'(\sigma(\lambda, \mu)^* f) = \sigma(\lambda, \mu)^* I'(f). \quad (1.2.2)$$

Put $h_{n,m}(t) = I'(x^n y^m)$. Taking $\mu = \lambda^{-1}$ in (1.2.2), one finds that $h_{n,m}(t) = \lambda^{n-m} h_{n,m}(t)$; hence $h_{n,m} = 0$ for $n \neq m$. By continuity and $k[[t]]$ -linearity of I' , for $f = \sum a_{n,m} x^n y^m$ one has

$$I'(f) = I' \left(\sum a_{n,m} x^n y^m \right) = I' \left(\sum a_{n,n} x^n y^n \right) = I'(\pi^* I(f)) = I(f) \cdot I'(1).$$

Taking $\mu = 1$ and $f = 1$ in (1.2.2), one obtains $h_{0,0}(\lambda t) = h_{0,0}(t)$, so that $I'(1) = h_{0,0}(t)$ is a constant c , and $I'(f) = cI(f)$. This constant is independent of n . To prove that $c = -1$, we may, and will, suppose $n = 1$.

The scheme $X'_1 = \text{Spec } k[[x, y]]/(xy)$ is the union of two nonsingular formal arcs crossing transversely, the first parametrized by y , the second by x . Sections of ω on X'_1 are systems of differential forms α and β on these arcs, with a simple pole at O , and $\text{Res } \alpha + \text{Res } \beta = 0$. Sections of ω on $X'_1 - \{O\}$ are systems of forms α and β with a pole of arbitrary order at O , and the composite

$$H^0(X'_1 - \{O\}, \omega) \longrightarrow H^1_{\{O\}}(X'_1, \omega) \xrightarrow{\text{Tr}} k$$

is the negative of $(\alpha, \beta) \mapsto \text{Res } \alpha + \text{Res } \beta$. With this notation, $dx \wedge dy/dt$ corresponds to

$$\left(\frac{dy}{y}, -\frac{dx}{x} \right),$$

and $(dx \wedge dy/dt)u_1 = \left(\frac{dy}{y}, 0 \right)$, whose sum of residues is 1. This proves (1.2.1).

2 Relation with Pic

2.1.

Let $S = \text{Spec } k[[t]]$, let s be the closed point of S , let η be the generic point, let $\pi : Y \rightarrow S$ be a proper flat morphism purely of relative dimension 1, and let O be a rational point of $Y_s := \pi^{-1}(s)$. Assume that, if X is the completion of Y at O , the morphism induced by π from X to S is of the type considered in Section 1. This means: Y is regular at O , O is an isolated point of non-smoothness of π , π has at O a non-degenerate quadratic singularity, and the two branches of Y_s at O are defined over k . An order between these branches is assumed to have been chosen.

It will be convenient to assume in addition that Y is regular and connected. Make the base change $\eta \rightarrow S$, and let Y_η be the curve over $k((t))$ which is the generic fibre of $\pi : Y \rightarrow S$.

2.2. Lemma.

The curve Y_η is absolutely irreducible and generically smooth over $k((t))$.

The curve Y_s is connected (because Y is proper over S and connected) and has a rational point, hence is absolutely connected. It is smooth near O (except at O), and therefore has rational smooth points over a separable extension k' of k . Over $S' = \operatorname{Spec} k'[[t]]$, $Y' = Y \times_S S'$ therefore has a section. The scheme Y' is connected, because Y'_s is connected, and regular, because it is étale over Y . It is therefore irreducible, and Y'_η is irreducible, hence connected. Having a rational point, it is absolutely connected. This shows that Y_η is absolutely connected. Being regular, it is absolutely irreducible. The smooth locus of Y_η , non-empty by hypothesis, is necessarily dense.

2.3. Interlude.

Let C be a complete reduced and generically smooth curve (i.e. C absolutely reduced) over a field K , and let D be a Cartier divisor on C . We will use the following geometric interpretation of $H^1(C, \mathcal{O}(-D))$.

For simplicity suppose C is connected and $D \neq 0$. This case will suffice for us. Let $\operatorname{Pic}(C; D)$ be the generalized Jacobian which parametrizes invertible sheaves on C trivialized on D . More precisely, define $\operatorname{Pic}(C; D)$ to be the K -scheme representing the functor on K -schemes

$$T \longmapsto \{\text{isomorphism classes of invertible sheaves on } C_T \text{ trivialized on } D_T\}.$$

Here $C_T = C \times_{\operatorname{Spec} K} T$, and similarly for D . It is a smooth K -scheme. Denote by $\mathcal{O}(1 \text{ on } D)^*$ the sheaf of sections of $\mathcal{O}^*$ whose restriction to D is 1. One has

$$\operatorname{Pic}(C; D)(T) = H^1(C_T, \mathcal{O}(1 \text{ on } D_T)^*).$$

Taking T to be the spectrum of the dual numbers over K , one obtains

$$\operatorname{Lie} \operatorname{Pic}(C; D) = H^1(C, \mathcal{O}(-D)). \quad (2.3.1)$$

Let ω be the dualizing sheaf on C ($\Omega_{C/K}^1$ for C smooth). We will use Serre duality between $H^0(C, \omega(D))$ and $H^1(C, \mathcal{O}(-D))$. It admits the following geometric interpretation, which we shall not use. Suppose first that C is smooth. For every T and every section $u : T \rightarrow C_T$ of C_T/T , the image of u is then a Cartier divisor on C_T . The invertible sheaf $\mathcal{O}(u(T))$ is canonically trivialized on the complement of $u(T)$. In particular, if u is a section of $(C - D)_T$, it is trivialized on D_T . This construction defines a morphism

$$j : C - D \longrightarrow \operatorname{Pic}(C; D), \quad u \longmapsto \text{class of } \mathcal{O}(u(T)). \quad (2.3.2)$$

Pullback by j gives a morphism from the vector space of invariant differential forms on $\operatorname{Pic}(C; D)$ - the dual of the Lie algebra - into $H^0(C - D, \omega)$. It gives Serre duality, i.e. the following diagram is commutative:

$$\begin{array}{ccc} (\operatorname{Lie} \operatorname{Pic}(C; D))^\vee & \xrightarrow{j^*} & H^0(C - D, \omega) \\ (2.3.1) \downarrow & & \uparrow \\ H^1(C, \mathcal{O}(-D))^\vee & \xrightarrow{\sim} & H^0(C, \omega(D)) \end{array}$$

For C only assumed absolutely reduced, with smooth locus U , the same arguments give $j : U - D \rightarrow \operatorname{Pic}(C; D)$. One still has (2.3.2), with $H^0(C - D, \omega)$ replaced by

$$H^0(U - D, \omega) = H^0(U - D, \Omega^1).$$

2.4.

Let Z be a relative Cartier divisor on Y , i.e. a divisor no component of which is contained in Y_s . Assume $O \notin Z$. The discussion in 2.3 will be applied to $K = k((t))$, $C = Y_\eta$, and $D = Z_\eta$.

Let $\text{Pic}(Y; Z)^0$ be the S -group scheme parametrizing invertible sheaves on Y , trivialized on Z , whose restriction to each irreducible component of each geometric fibre has degree 0. For the existence of Pic , see [2], 8.2. It is a smooth group scheme over S , with connected fibres. Its special fibre $\text{Pic}(Y; Z)_s^0$ is the identity component of the group scheme $\text{Pic}(Y_s; Z_s)$ classifying invertible sheaves on Y_s trivialized on Z_s .

Let Y'_s be the curve obtained from Y_s by normalizing at O : we have $h : Y'_s \rightarrow Y_s$, and $h^{-1}(O)$ consists of two points O_1 and O_2 corresponding to the two branches of Y_s at O ; Y_s is obtained from Y'_s by identifying O_1 and O_2 . In particular, h induces an isomorphism from $Y'_s - \{O_1, O_2\}$ to $Y_s - \{O\}$. By abuse of notation, we again denote by Z_s the inverse image of Z_s in Y'_s . The functor h^* identifies invertible sheaves $\mathcal{L}$ on Y_s with invertible sheaves $\mathcal{L}'$ on Y'_s , equipped with an isomorphism

$$\sigma : \mathcal{L}'_{O_1} \xrightarrow{\sim} \mathcal{L}'_{O_2}$$

between the fibres at O_1 and O_2 . A trivialization of $\mathcal{L}$ on Z_s corresponds to one of $\mathcal{L}'$. In particular, for $\lambda \in k^*$ one obtains an invertible sheaf $\mathcal{L}_\lambda$ on Y_s , trivialized on Z_s , by taking $\mathcal{L}' = \mathcal{O}$ and taking $\sigma : k \rightarrow k$ to be multiplication by λ . The same construction can be repeated after a base change $T \rightarrow s$, whence a morphism of k -group schemes

$$w_1 : \mathbb{G}_m \longrightarrow \text{Pic}(Y_s, Z_s), \quad \lambda \longmapsto \mathcal{L}_\lambda. \quad (2.4.1)$$

Here is an analogous construction modulo t^n . Let Y_n be the reduction of Y modulo t^n , and let $q : Y''_n \rightarrow Y_n$ coincide with Y_n away from O and, in a neighbourhood of O , be obtained as follows: taking “neighbourhood” in the étale sense, one can find local coordinates x, y in which Y_n is written $(xy)^n = 0$. We take Y''_n to be the disjoint union of $x^n = 0$ and $y^n = 0$. Note that for $n > 1$, Y''_n is not flat over $k[[t]]/(t^n)$. A local calculation shows that $\mathcal{O}_{Y_n}$ injects into $q_*\mathcal{O}_{Y''_n}$. This remains true after any flat base change $T \rightarrow \text{Spec } k[[t]]/(t^n)$. Define $Q_{n,T}$ by the exact sequence

$$0 \longrightarrow \mathcal{O}_{Y_{n,T}} \longrightarrow q_*\mathcal{O}_{Y''_{n,T}} \longrightarrow Q_{n,T} \longrightarrow 0. \quad (2.4.2)$$

This is a sheaf supported on the inverse image O_T of O in $Y_{n,T}$. Writing $\pi^\bullet$ for inverse image in the sheaf-theoretic sense by $\pi : Y_{n,T} \rightarrow T$, we have a morphism of sheaves

$$\delta : (\pi^\bullet \mathcal{O}_T)_{|O_T} \longrightarrow Q_{n,T} \quad (2.4.3)$$

which, locally near O for the étale topology, sends u to the image in $q_*\mathcal{O}_{Y''_{n,T}}/\mathcal{O}_{Y_{n,T}}$ of the section of $\mathcal{O}_{Y''_{n,T}}$ which is u on the first branch and 0 on the second.

Passing to the multiplicative group, define $Q_{n,T}^*$ similarly by the exact sequence

$$0 \longrightarrow \mathcal{O}_{Y_{n,T}}^* \longrightarrow q_*\mathcal{O}_{Y''_{n,T}}^* \longrightarrow Q_{n,T}^* \longrightarrow 0, \quad (2.4.4)$$

and

$$\delta^* : (\pi^\bullet \mathcal{O}_T^*)|_{\mathcal{O}_T} \longrightarrow Q_{n,T}^*. \quad (2.4.5)$$

Here $\delta^*(u)$ is the image of “ u on the first branch, 1 on the second”.

Taking the boundary $H^0 \rightarrow H^1$ in the long exact cohomology sequence with support in $\mathcal{O}_T$ deduced from (2.4.2), respectively (2.4.4), one deduces from (2.4.3), respectively (2.4.5), the respective morphisms

$$H^0(T, \mathcal{O}_T) \longrightarrow H_{\mathcal{O}_T}^1(Y_{n,T}, \mathcal{O}) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(-Z_{n,T})) \quad (2.4.6)$$

$$H^0(T, \mathcal{O}_T^*) \longrightarrow H_{\mathcal{O}_T}^1(Y_{n,T}, \mathcal{O}^*) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(1 \text{ on } Z_{n,T})^*). \quad (2.4.7)$$

The morphism of functors in T given by (2.4.7) supplies a morphism of $k[[t]]/(t^n)$ -schemes from $\mathbb{G}_m$ into the reduction modulo t^n of $\text{Pic}_Z(Y)^0$:

$$w_n : \mathbb{G}_m \longrightarrow \text{Pic}(Y, Z)_n^0. \quad (2.4.8)$$

For $m \leq n$, w_n has reduction modulo t^m equal to w_m , and for $n = 1$ one recovers (2.4.1). By SGA3 IX 3.6, there is a unique morphism (2.4.8) lifting the k -morphism (2.4.1) to $k[[t]]/(t^n)$. We will use the fact that it has the description (2.4.7).

One cannot pass to the limit over n to deduce from the w_n a morphism of group schemes from $\mathbb{G}_m$ to $\text{Pic}(Y; Z)^0$. However:

- (a) for H a finite scheme over $k[[t]]$ and P arbitrary, giving a morphism $H \rightarrow P$ is equivalent to giving a coherent system of morphisms $H_n \rightarrow P_n$ between the reductions modulo t^n ;
- (b) for V and W two free $k[[t]]$ -modules, giving a linear map $F : V \rightarrow W$ is equivalent to giving a coherent system of linear maps $V \otimes k[[t]]/(t^n) \rightarrow W \otimes k[[t]]/(t^n)$.

For every integer m , one may apply (a) to the subgroup scheme μ_m of $\mathbb{G}_m$, the kernel of $x \mapsto x^m : \mathbb{G}_m \rightarrow \mathbb{G}_m$: passing to the limit over n , the w_n give a morphism

$$w : \mu_m \longrightarrow \text{Pic}(Y, Z)^0. \quad (2.4.9)$$

One may apply (b) after passage to Lie algebras: one obtains

$$dw : \text{Lie } \mathbb{G}_m \longrightarrow \text{Lie } \text{Pic}(Y; Z)^0. \quad (2.4.10)$$

We have $\text{Lie } \mathbb{G}_m = \mathcal{O}$, the trivial free module of rank one, and the arguments of 2.3 again give a canonical isomorphism $\text{Lie } \text{Pic}(Y; Z)^0 = H^1(Y, \mathcal{O}(-Z))$. Thus the datum of dw is equivalent to that of the image

$$dw(1) \in H^1(Y, \mathcal{O}(-Z)). \quad (2.4.11)$$

Let $\omega_{Y/S} = \pi^! \mathcal{O}_S[1]$ be the relative dualizing sheaf, and let φ be a section of $\omega_{Y/S}(Z)$. Choose on X , the completion of Y at $\mathcal{O}$, formal coordinates x, y in which π is $(x, y) \mapsto xy$, and in which the first branch of X_1 is $x = 0$. In this coordinate system, φ is written on X as

$$\varphi = f(x, y) dx \wedge dy/dt.$$

The restriction of φ_η to Y_η belongs to the $k((t))$ -vector space $H^0(Y_\eta, \omega(Z_\eta))$, in Serre duality with $H^1(Y_\eta, \mathcal{O}(-Z_\eta))$. Our aim in this paragraph is the following:

2.5. Proposition.

With the preceding notation, one has

$$I(f) = -\langle dw(1), \varphi \rangle. \quad (2.5.1)$$

Over all of S one still has a pairing between $\text{Lie Pic}(Y; Z)^0 = H^1(Y, \mathcal{O}(-Z))$ and $H^0(Y, \omega(Z))$. Under the hypotheses made, these $k[[t]]$ -modules are in fact free, and the pairing is a perfect duality. The right-hand side of (2.5.1) is therefore in $k[[t]]$, and not merely in $k((t))$. Its reduction modulo t^n is $\langle dw_n(1), \varphi \rangle$. We verify (2.5.1) modulo t^n for each n .

Let X_n and Y_n be the reductions of X and Y modulo t^n . The local trace morphism $H^1_{\{O\}}(X_n, \omega) \rightarrow k[[t]]/(t^n)$ and the global trace morphism $H^1(Y_n, \omega) \rightarrow k[[t]]/(t^n)$, which gives Serre duality, are related by the commutative diagram

$$\begin{array}{ccccc} H^1_{\{O\}}(X_n, \omega) & \longleftarrow & H^1_{\{O\}}(Y_n, \omega) & \longrightarrow & H^1(Y_n, \omega) \\ \uparrow & & & & \uparrow \\ k[[t]]/(t^n) & \xlongequal{\hspace{1.5cm}} & & & k[[t]]/(t^n) \end{array}$$

This compatibility reduces 2.5.1 to the lemma.

2.6. Lemma.

The morphism

$$H^1_{\{O\}}(X_n, \mathcal{O}) \longleftarrow H^1_{\{O\}}(Y_n, \mathcal{O}(-Z_n)) \longrightarrow H^1(Y_n, \mathcal{O}(-Z_n))$$

sends v_n to $dw_n(1)$.

We leave to the reader the verification that the image of v_n is also the image of 1 by (2.4.6). The lemma then follows from the description of dw_n in terms of the dual numbers $D = (k[[t]]/(t^n))[\varepsilon]/(\varepsilon^2)$ over $k[[t]]/(t^n)$ and from the natural exact sequence of morphisms

$$((2.4.6) \text{ over } k[[t]]/(t^n)) \longrightarrow ((2.4.7) \text{ over } D) \longrightarrow ((2.4.7) \text{ over } k[[t]]/(t^n)).$$

3 Proof of the theorem

3.1.

Let S_0 be a smooth curve over k (a smooth scheme purely of dimension 1), let s be a rational point of S_0 , and let S be the completion of S_0 at s . Let $\pi : Y_0 \rightarrow S_0$ be proper and flat, purely of relative dimension 1, let Y be the inverse image of Y_0 on S , and let $O \in Y_s$. We again denote by π the morphism $Y \rightarrow S$ deduced from $\pi : Y_0 \rightarrow S_0$, and assume that $(\pi : Y \rightarrow S, O)$ is of the type considered in §2. We use the notation of Section 2 for $(\pi : Y \rightarrow S, O)$.

Finally assume, for simplicity, that Y_0 is regular and that π has geometrically connected fibres. Let $\omega := \omega_{Y_0/S_0}$ be the relative dualizing sheaf. Where Y_0 is smooth over S_0 , this is $\Omega^1_{Y_0/S_0}$. We again write ω for its inverse image on Y , or

on the completion X of Y at O . We choose formal coordinates on X and S as before.

Let φ be a rational section of ω on Y_0 , regular at O . On X , one can write $\varphi = f(x, y)dx \wedge dy/dt$. The main result is the following:

3.2. Proposition.

If k has characteristic $p > 0$, then $I(f)$ is algebraic over the field $k(S_0) \subset k((t))$ of rational functions on S_0 .

After replacing φ by its product with a rational function on S_0 vanishing sufficiently at s , and after replacing S_0 by a Zariski open neighbourhood containing s , we may and do suppose that, for a suitable relative Cartier divisor Z_0 on Y_0 not containing O , φ is a section of ω on $Y_0 - Z_0$.

Let $\text{Pic}(Y_0; Z_0)^0$ be the group scheme over S_0 which parametrizes invertible sheaves on Y_0 , trivialized on Z_0 , of degree 0 on each irreducible component of each geometric fibre; if necessary to apply [2], shrink S_0 further. We interpret φ as corresponding to a section φ' of the dual of the vector bundle $\text{Lie Pic}(Y_0; Z_0) = R^1\pi_*\mathcal{O}(-Z_0)$ on S_0 by Serre duality; in the geometric interpretation (2.3) one has $\varphi = j^*\varphi'$.

After the base change from S_0 to S , we have $dw : \mathcal{O} \rightarrow \text{Lie Pic}(Y_0, Z_0)_S^0 = \text{Lie Pic}(Y; Z)^0$ (2.4.10), and the proposition follows from 2.5 and the key lemma.

3.3. Key Lemma.

If $p > 0$, then dw is algebraic over S_0 .

In characteristic $p > 0$, the inclusion $\mu_p \subset \mathbb{G}_m$ induces an isomorphism on Lie algebras:

$$\text{Lie}(\mu_p) \xrightarrow{\sim} \text{Lie}(\mathbb{G}_m).$$

By reduction modulo t^n , one verifies that dw is the differential of the morphism (2.4.9) $w : \mu_p \rightarrow \text{Pic}_Z(Y)^0$. It remains only to use the rigidity of μ_p : for every flat commutative group scheme P over S_0 , $\text{Hom}(\mu_p, P)$ is represented by an étale scheme over S_0 . Take $P = \text{Pic}(Y_0; Z_0)$, and let $\tilde{S}_0$ be the resulting S_0 -scheme. By definition, over $\tilde{S}_0$ there is a universal morphism

$$\text{un} : \mu_p \longrightarrow \text{Pic}(Y_0; Z_0)_{\tilde{S}_0}. \quad (3.3.1)$$

In particular, the morphism 2.4.9 is the inverse image of 3.3.1 by an S_0 -morphism b :

$$\begin{array}{ccc} S & \xrightarrow{b} & \tilde{S}_0 \\ & \searrow & \swarrow \\ & S_0 & \end{array}$$

Likewise, dw is the inverse image by b of the morphism from $\mathcal{O}$ to $\text{Lie Pic}(Y_0; Z_0)_{\tilde{S}_0}$ deduced by passage to Lie algebras from the homomorphism (3.3.1), and, since $\tilde{S}_0$ is quasi-finite over S_0 , this proves its algebraicity.

3.4.

Let $k\{x_1, \dots, x_N\}$ denote the henselization of the local ring of affine space $\text{Spec } k[x_1, \dots, x_N]$ at 0. The completion of this local ring is $k[[x_1, \dots, x_N]]$, and it follows from [3] XI §3, Cor. 1, that for $g \in k[[x_1, \dots, x_N]]$ to be algebraic over the field $k(x_1, \dots, x_N)$, it is necessary and sufficient that g belong to $k\{x_1, \dots, x_N\}$.

3.5. Proof of the theorem.

By 3.4, saying that f is algebraic means that there exist a surface Y'_0 , with a rational point O , an étale morphism h sending O to 0 from Y'_0 to the plane $\mathbb{A}^2$ with coordinates x, y , and a section $\tilde{f}$ of $\mathcal{O}$ on Y'_0 whose expansion in a power series at O is f . Let S'_0 be the line $\mathbb{A}^1$ with coordinate t , and let π' be the composite morphism

$$\pi' : Y'_0 \xrightarrow{h} \mathbb{A}^2 \xrightarrow{xy} S'_0.$$

On Y'_0 , let φ be the section $\tilde{f} \cdot h^*(dx \wedge dy/dt)$ of the relative dualizing sheaf. Compactifying Y'_0 and resolving the singularities introduced, one obtains

$$\begin{array}{ccc} Y'_0 & \longrightarrow & Y_0 \\ \pi' \downarrow & & \downarrow \pi'' \\ S'_0 & \xlongequal{\quad} & S'_0 \end{array}$$

with π'' proper and flat and Y_0 regular. On Y_0 , φ appears as a rational section of the relative dualizing sheaf. In order that π have geometrically connected fibres, replace S'_0 by the space S_0 of the Stein factorization

$$Y_0 \xrightarrow{\pi} S_0 \longrightarrow S'_0$$

of π'' . It remains only to apply 3.2 to φ .

3.6. Proof of the corollary.

We proceed by induction on N . The case $N = 0$ is left to the reader, the case $N = 1$ is trivial, the case $N = 2$ is the theorem, and we suppose $N \geq 3$. For g in $k[[x_1, \dots, x_N]]$, put

$$g = \sum g_{nm} x_{N-1}^n x_N^m$$

with $g_{nm} \in k[[x_1, \dots, x_{N-2}]]$, and

$$I(g) = \sum g_{nn} x_{N-1}^n.$$

One has

$$I_N(g) = I_{N-1}I(g). \tag{3.6.1}$$

If g is algebraic, then $g \in k\{x_1, \dots, x_N\}$ (3.4), and the g_{nm} are algebraic: $g_{nm} \in k\{x_1, \dots, x_{N-2}\}$. Let k' be the field of fractions of $k\{x_1, \dots, x_{N-2}\}$, and consider the commutative diagram

$$\begin{array}{ccccc} k[[x_1, \dots, x_N]] & \longrightarrow & k\{x_1, \dots, x_{N-2}\}[[x_{N-1}, x_N]] & \hookrightarrow & k'[[x_{N-1}, x_N]] \\ I \downarrow & & \downarrow I & & \downarrow I \\ k[[x_1, \dots, x_{N-1}]] & \longrightarrow & k\{x_1, \dots, x_{N-2}\}[[x_{N-1}]] & \hookrightarrow & k'[[x_{N-1}]]. \end{array}$$

In the first row these are $k[x_1, \dots, x_N]$ -algebras, and in the second row $k[x_1, \dots, x_{N-1}]$ -algebras. Apply the theorem to the image g' of g , assumed algebraic, in $k'[[x_{N-1}, x_N]]$. Since k' is algebraic over the field of fractions of $k[x_1, \dots, x_{N-2}]$, the series $I(g')$, algebraic over $k'(x_{N-1})$, satisfies a nontrivial equation

$$\sum a_i I(g')^i = 0$$

with coefficients $a_i \in k[x_1, \dots, x_{N-1}]$. The same equation is satisfied by $I(g)$, hence $I(g)$ is algebraic: $I(g) \in k\{x_1, \dots, x_{N-1}\}$. We conclude by (3.6.1) and the induction hypothesis.

3.7.

One can extract from the proof of the theorem an estimate for the degree d of the equation expressing that $I(f)$ is algebraic over $k(t)$. With the notation of 3.5 and 3.1, suppose that S_0 has degree σ over S'_0 , that Y_η has genus g , and that Z_η has z' geometric points. Put $z = \sup(0, z' - 1)$. The genus here is the genus of the normalization of the curve $Y_{\bar{\eta}}$ deduced from Y_η by extension of scalars to an algebraic closure $k(\bar{\eta})$ of $k(\eta)$; the first l -adic Betti number of $Y_{\bar{\eta}} - Z_{\bar{\eta}}$ is $2g + z$. One has

$$d \leq \sigma \cdot p^{g+z}. \quad (3.7.1)$$

The degree of $I(f)$ over $k(S_0)$ is $\leq p^{g+z}$, and even $\leq \sup((p^g - 1)p^z, z'!)$.

If f is an algebraic formal power series with integral coefficients and if σ , g , and z are defined in terms of $f \in \mathbb{Q}[[x, y]]$, one deduces that the degree d_p of the reduction modulo p of $I(f)$ satisfies, for almost all p ,

$$d_p \leq \sigma \cdot p^{g+z}. \quad (3.7.2)$$

One can also obtain information modulo p^m : using the fact that over a base where $p^m = 0$ one has $\text{Lie}(\mu_{p^m}) \xrightarrow{\sim} \text{Lie}(\mathbb{G}_m)$, one finds that, for almost all p , the reduction of $I(f)$ modulo p^m satisfies an equation with coefficients in $\mathbb{Z}/(p^m)[t]$, not all divisible by p , and of degree $\leq \sigma \cdot p^{m(g+z)}$.

3.8.

I do not know how to obtain information modulo p^m for every p . If in the proof of the theorem one tries to replace k by a commutative ring A in which $p^m = 0$, a difficulty appears: that of constructing Y_0/S_0 not too badly behaved, for example flat, not only near O but everywhere.

3.9.

The proof in 3.6 of the corollary does not satisfy me. It would be more interesting to treat the case of n variables directly, inspired by the following analogue of the integral formula (1): over $\mathbb{C}$, for g convergent, one has

$$I(g)(t) = \int_{Z(t)} g dz_1 \cdots dz_n / dt, \quad (1')$$

where the integral is over a cycle

$$Z(t) = \{(x_1, \dots, x_n) \mid x_1 \cdots x_n = t, |x_1| = r_1, \dots, |x_n| = r_n\}$$

with $\prod r_i = |t|$.

I hope that a direct proof would give, in the case of algebraic formal power series with integral coefficients, an estimate $d_p \leq O(p^n)$ for the degree of the reduction of $I(g)$ modulo p over $\mathbb{F}_p(t)$.

To provide the tools required for a direct proof would be a test for a p -adic theory of vanishing cycles.

3.10.

Furstenberg's proof of the corollary, for rational g , shows that if $g \in k[[x_1, \dots, x_N]]$ is one of the components of a column vector h of elements of $k[[x_1, \dots, x_N]]$ satisfying a system of equations

$$h = Ah^p$$

for A a square matrix with polynomial coefficients, then $I(g)$ has the same property, and is therefore algebraic. This kind of equation is reminiscent of F -crystals in crystalline cohomology.

In parallel, $T_p \text{Pic}$ is an avatar of p -adic H^1 . This leads me to hope that Furstenberg's proof and the proof given here are less distant from one another than they seem.

Appendix

Signs

The treatment in the literature of sign questions in the duality formalism for coherent algebraic sheaves has seemed confused to me. Hartshorne's text *Residues and Duality* (Lect. Notes 20) suffers from the following defects:

- (a) the convention defining the complexes $\text{Hom}^\bullet(K, L)$ (I 6, pp. 63, 64) and $K \otimes L$ (II 4, p. 93) is not the usual convention; contrary to what is asserted on p. 64, $\text{Hom}^\bullet$ does not commute identically with translation functors (cf. SGA4 XVII 0.3); the same is true for $\otimes$, whose associativity requires the definition of a canonical isomorphism, which is not given. Since the definition of $f^!$ for $f : X \rightarrow Y$ smooth purely of relative dimension n involves the tensor product with $\Omega_{X/Y}^n[n]$, this lacuna matters.
- (b) As far as I can nevertheless understand the proposed convention, the definition III 4.3 of the trace morphism, based on the convention which in III 3.4 defines γ , seems to me incompatible with III 10.5 TRA1).

The conventions set out below are, I believe, coherent. I use the usual sign convention for $\otimes$ (SGA4 XVII 1.1). I have not restricted myself to dimension 1, which would suffice for the needs of the article.

(a)

For $f : X \rightarrow S$ flat of relative complete-intersection type, purely of relative dimension n , the relative dualizing sheaf $\omega_{X/S}$ is related to the relative dualizing complex $K_{X/S} := Rf^! \mathcal{O}_S$ by

$$K_{X/S} = \omega_{X/S}[n]. \quad (\text{S.1})$$

It is an invertible sheaf. For X smooth over S , one has $\omega_{X/S} = \Omega_{X/S}^n$. If, as suggested in my appendix to Hartshorne's book, one defines $Rf^!$ by adjunction, the equality $\omega_{X/S} = \Omega_{X/S}^n$ is a canonical isomorphism whose definition poses a sign problem. We adopt here the opposite point of view (cf. Hartshorne's treatment of smoothable morphisms), where this equality is evident from the definition of $Rf^!$ and the sign problem lies in the definition of the trace morphism (S.5) below.

If X is the fibre product over S of the $f_i : X_i \rightarrow S$ ($i \in I$), where each X_i is flat of relative complete-intersection type and purely of relative dimension n_i , with $n = \sum n_i$, one has

$$K_{X/S} = \boxtimes_i K_{X_i/S} \quad (\text{S.2})$$

(external tensor product). By definition, the datum of such an isomorphism is the datum, for each total order on I , of an isomorphism

$$\omega_{X/S} = \bigotimes_i \omega_{X_i/S}. \quad (\text{S.3})$$

We require that when one passes from the total order $<_1$ to the total order $<_2$, the isomorphism change by the sign $(-1)^N$, where $N = \sum n_i n_j$, the sum being over the (i, j) with $i <_1 j$ and $i >_2 j$. For X_i smooth over S , one has $\omega_{X_i/S} = \Omega_{X_i/S}^{n_i}$, and (S.3) is given by exterior product, taken in the given order on I . It has the desired parity property.

For an iterated morphism $f = gh : X \xrightarrow{h} Y \xrightarrow{g} S$, $K_{X/S}$ is the tensor product of the complexes $K_{X/Y}$ and $h^* K_{Y/S}$ by an isomorphism which, applied to $X_1 \times_S X_2 \rightarrow X_1 \rightarrow S$, gives back (S.3).

(b)

For $f : X \rightarrow S$ as in (a) and proper, the trace morphism is a morphism

$$\text{Tr} : Rf_* K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.4})$$

The morphism (S.4) induces a morphism of coherent sheaves $\mathcal{H}^0 Rf_* K_{X/S} \rightarrow \mathcal{O}_S$, i.e. a morphism

$$\text{Tr} : R^n f_* \omega_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.5})$$

Let us specify how one passes from (S.4) to (S.5). For every complex K , the complex $K[d]$ has components $(K[d])^i = K^{i+d}$ and differential that of K modified by the sign $(-1)^d$. We identify the $\mathcal{H}^i(K[d])$ with $\mathcal{H}^{i+d} K$ without introducing a sign. The functor Rf_* commutes with the translation functor $[d]$, and (S.5) is deduced from (S.4) by applying the functor $\mathcal{H}^0$: use the isomorphism

$$\begin{aligned} \mathcal{H}^0 Rf_* K_{X/S} &= \mathcal{H}^0 Rf_*(\omega_{X/S}[n]) = \mathcal{H}^0(Rf_*(\omega_{X/S})[n]) \\ &= \mathcal{H}^n Rf_* \omega_{X/S} = R^n f_* \omega_{X/S}. \end{aligned}$$

For a section s of f , there is also a local trace morphism

$$\mathrm{Tr} : Rs^!K_{X/S} \longrightarrow \mathcal{O}_S. \quad (\text{S.6})$$

The definitions of (S.4) and (S.6) pose, in each relative dimension n , a sign problem. We impose commutativity of the diagram

$$\begin{array}{ccc} Rs^!K_{X/S} & \longrightarrow & Rf_*K_{X/S} \\ \downarrow & & \downarrow \\ \mathcal{O}_S & \xlongequal{\quad} & \mathcal{O}_S, \end{array}$$

which relates, in each relative dimension, the choices (S.4) and (S.6). The morphism in the first row of the diagram is derived from the inclusion $s^! \rightarrow f_*$, “sections with schematic support in $s(S)$ ” included in “all sections”. For X_i as in (a), proper over S , and hence giving by (S.3) an isomorphism

$$Rf_*K_{X/S} = \bigotimes_i Rf_{i*}K_{X_i/S},$$

we impose that the morphism (S.4) for X/S be the tensor product of the morphisms (S.4) for the X_i/S . This relates the conventions used in each relative dimension and leaves only one sign ambiguity. We resolve it by requiring that, for X a smooth curve over an algebraically closed field k and $x \in X(k)$, the local trace morphism

$$H_{\{x\}}^0(X, \Omega_X^1[1]) \longrightarrow k, \quad (\text{S.7})$$

which induces (S.6) for s the inclusion of x , be the following. The complex $\Omega_X^1[1]$, reduced to the sheaf Ω_X^1 in cohomological degree -1 , maps quasi-isomorphically to the complex reduced to degrees -1 and 0

$$L := \Omega_X^1(\infty \cdot x) \longrightarrow \Omega_X^1(\infty \cdot x)/\Omega_X^1.$$

One has

$$H_{\{x\}}^0(X, \Omega_X^1[1]) \xrightarrow{\sim} H_{\{x\}}^0(X, L) \xleftarrow{\sim} H_{\{x\}}^0(X, L) = (\Omega_X^1(\infty \cdot x)/\Omega_X^1)_x,$$

the polar parts at x of differential forms, and for the local trace morphism we take the residue morphism Res_x .

Let us indicate a few compatibilities to which this choice of signs gives rise. To simplify notation, we take S to be the spectrum of a field k .

(c)

Let X be a smooth curve over an algebraically closed field k , and let $x \in X(k)$. Again call “trace” the morphism deduced from (S.7)

$$\mathrm{Tr} : H_{\{x\}}^1(X, \Omega_X^1) \longrightarrow k.$$

Let L' be the complex in degrees 0 and 1

$$L' := \Omega_X^1(\infty \cdot x) \longrightarrow \Omega_X^1(\infty \cdot x)/\Omega_X^1.$$

It differs by a sign from $L[-1]$. Because of this sign, Tr is the negative of the composite

$$H_{\{x\}}^1(X, \Omega_X^1) \xleftarrow{\sim} H_{\{x\}}^1 L' = (\text{polar parts at } x) \xrightarrow{\text{Res}} k.$$

The sequence of functors $\Gamma_{\{x\}}(X, -) \rightarrow \Gamma(X, -) \rightarrow \Gamma(X - \{x\}, -)$, which is exact short when applied to a flasque sheaf, gives a boundary morphism $\partial : H^0(X - \{x\}, \Omega^1) \rightarrow H_{\{x\}}^1(X, \Omega^1)$. One has

$$\text{Tr} \circ \partial = -\text{Res}_x.$$

Finally, for X complete, the boundary morphism attached to the short exact sequence of sheaves

$$0 \longrightarrow \Omega_X^1 \longrightarrow (\text{rational differential forms}) \longrightarrow (\text{polar parts}) \longrightarrow 0,$$

composed with the trace morphism, is the sum of the residue morphisms at all points $x \in X(k)$.

(d)

For $X = X_1 \times X_2$, with X_i complete locally of complete-intersection type and purely of dimension n_i , and for $u_i \in H^{n_i}(X_i, \omega_{X_i})$, let $n = n_1 + n_2$ and let $\text{pr}_1^* u_1 \cup \text{pr}_2^* u_2 \in H^n(X, \omega_X)$ be the cup product relative to (S.3): $\text{pr}_1^* \omega_{X_1} \otimes \text{pr}_2^* \omega_{X_2} \xrightarrow{\sim} \omega_X$. The required compatibility of the trace morphisms with products is

$$\text{Tr}(\text{pr}_1^* u_1 \cup \text{pr}_2^* u_2) = (-1)^{n_1 n_2} \text{Tr}(u_1) \text{Tr}(u_2).$$

(e)

For $k = \mathbb{C}$, and for the transcendental topology, one has a trace morphism

$$\int : H^{2n}(X(\mathbb{C}), \mathbb{C}) \longrightarrow \mathbb{C}.$$

If Ω^{**} is the C^∞ de Rham complex, the quasi-isomorphism $\mathbb{C} \rightarrow \Omega^{**}$ gives an isomorphism $H^{2n}(X(\mathbb{C}), \mathbb{C}) = (2n\text{-forms})/(\text{exact forms})$, and $\int$ is integration over $X(\mathbb{C})$ with its usual orientation. It will be more convenient for us to regard $\int$ as a morphism $H^0(X(\mathbb{C}), \mathbb{C}[2n]) \rightarrow \mathbb{C}$. This even shift in the degrees causes no sign problem. Let Ω^* be the holomorphic de Rham complex. The quasi-isomorphisms and morphisms

$$\mathbb{C}[2n] \longrightarrow \Omega^*[2n] \longleftarrow \Omega^n[n]$$

give in cohomology, for the transcendental topology,

$$e : H^0(X(\mathbb{C}), \Omega^n[n]) \longrightarrow H^0(X(\mathbb{C}), \mathbb{C}[2n]).$$

One has $H^0(X(\mathbb{C}), \Omega^n[n]) = H^0(X, \Omega^n[n])$ by GAGA, and between the two trace morphisms the relation

$$\text{Tr}(u) = \frac{1}{(2\pi i)^n} \int e(u).$$

(f)

Let $\mathbb{P}^n$ be projective n -space, let $(T_i)_{0 \leq i \leq n}$ be a system of homogeneous coordinates, let U_i be the open set $T_i \neq 0$, and set $t_i = T_i/T_0$. Order the index set $[0, n]$ of the open covering (U_i) in the evident way. The section

$$\frac{dt_1}{t_1} \wedge \cdots \wedge \frac{dt_n}{t_n}$$

of Ω^n on the intersection $U_{0\dots n}$ of the U_i defines a Čech cocycle, and hence a class in $H^n(\mathbb{P}^n, \Omega^n)$. One has

$$\mathrm{Tr} \left(\frac{dt_1}{t_1} \wedge \cdots \wedge \frac{dt_n}{t_n} \right) = (-1)^{n(n+1)/2}.$$

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